55 A+ mid. 25 → -1.7

NATIONAL INSTITUTE OF TECHNOLGY PATNA

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

End Semester Examination Jan – Jun 2025

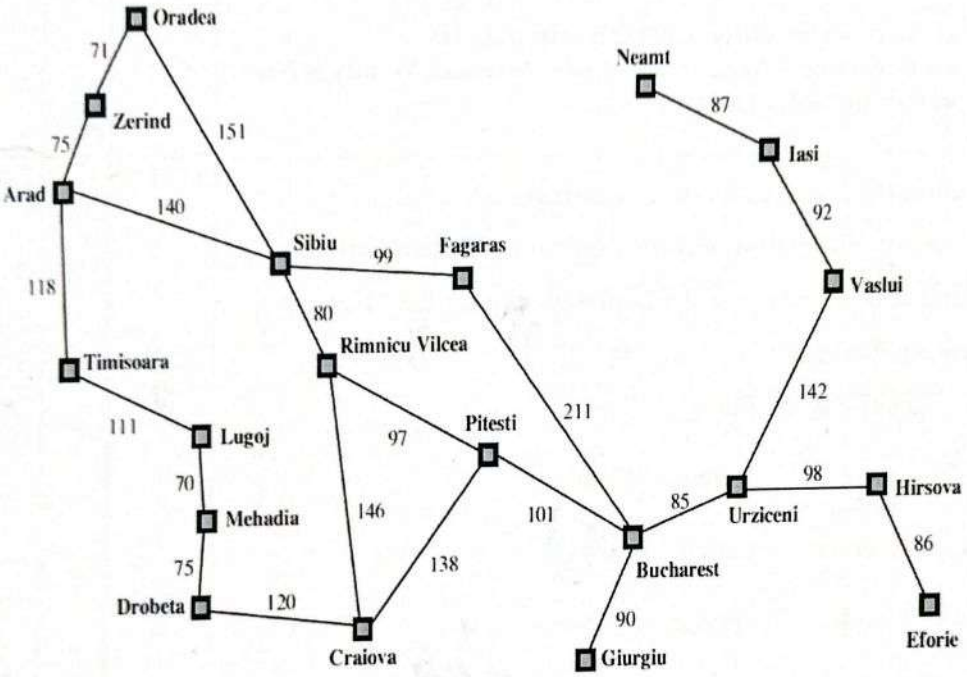
B. Tech - CSE

Course Name: Artificial Intelligence *Course Code: CS44112*

Maximum Time: 3 Hours **Max. Marks:60**

The paper contains three questions. Answer all questions and its parts at the same place.

S. N.	Question	Marks	CO	BL
1	Implement a Genetic Algorithm (GA) to find the **minimum value** of the function f(x)=x^3+5 where x is in the range [−10, 10].	[10]	CO2	Understand & Apply
2	Trace the operation of A* search applied to the problem of getting to Bucharest from Lugoj using the straight-line distance heuristic. Show the sequence of nodes that the algorithm will consider and the *f(n)*, *g(n)*, and *h(n)* score for each node.	[10]	CO3	Remember, Evaluate
3	a) Compare and contrast Forward State Space Search and Backward State Space Search approaches for planning, with suitable diagrams. b) Explain STRIPS in detail. Write a STRIPS representation for a *Robot moving blocks* **OR** *Block world problem* from one table to another.	[10]	CO4	Remember, Understand Remember & Understand



Arad	366	**Mehadia**	241
Bucharest	0	**Neamt**	234
Craiova	160	**Oradea**	380
Drobeta	242	**Pitesti**	100
Eforie	161	**Rimnicu Vilcea**	193
Fagaras	176	**Sibiu**	253
Giurgiu	77	**Timisoara**	329
Hirsova	151	**Urziceni**	80
Iasi	226	**Vaslui**	199
Lugoj	244	**Zerind**	374

Values of h_{SLD}—straight-line distances to Bucharest.

Shot on OnePlus

4	Given the following dataset w. r. t. Naïve Bayes algorithm, compute the prior probabilities for the classes (Play = Yes/No): **For the above dataset, predict whether a person will play if:** **Outlook = Sunny, Temperature = Mild, Humidity = Normal, Windy = No** (Show step-by-step probability calculations.)	[10]	CO5	Apply
5	a) What is the role of entropy in decision tree construction? b) How does Gini impurity differ from information gain in decision trees? c) Given the following dataset, compute the Gini impurity for the "Tip" class (High/Low) based on "Service": d) Construct a decision tree (up to 2 levels) to predict "Tip" using the above dataset. Justify your splits using entropy or Gini index.	[10]	CO5	Remember, Understand
6	a) A simple neural network has one input layer with 3 neurons, one hidden layer with 2 neurons, and one output neuron. How many total weights are there (including biases)? b) A single-layer perceptron has two inputs x_1 and x_2 with weights $w_1=0.4$ and $w_2 = -0.7$ and bias $b= 0.2$. The activation function is a **step function**: output is 1 if net input ≥ 0, else 0. Find the output for input vector $(x_1, x_2) = (1,1)$. c) Design a perceptron (show weights and bias) that implements the NAND gate. Justify your choices.	[10]	CO1	Create Creates Design

Outlook	Temperature	Humidity	Windy	Play
Sunny	Hot	High	No	No
Overcast	Hot	High	Yes	Yes
Rainy	Mild	High	No	Yes
Sunny	Cool	Normal	No	Yes
Rainy	Cool	Normal	Yes	No

Bill Amount	Service	Time	Tip
50	Good	Dinner	High
30	Average	Lunch	Low
60	Good	Dinner	High
20	Poor	Lunch	Low
70	Excellent	Dinner	High

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Department of Computer Science and Engineering

END SEMESTER EXAMINATION, Jan-June 2024

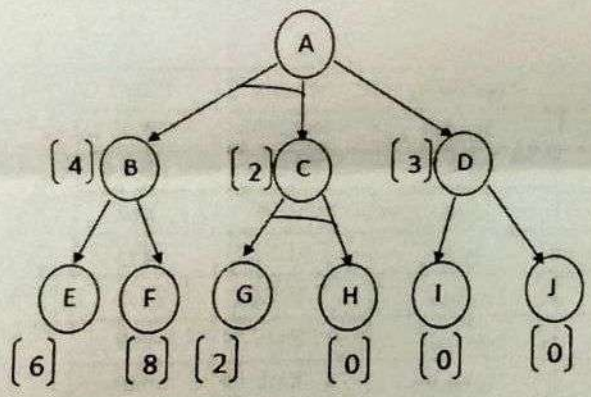
B. Tech: Semester-4 (Section I, II, III, Dual-Degree)

Course Name: Artificial Intelligence | **Course Code: CS4#112**

Maximum Time: 3 hours | **Max. Marks: 60**

Instructions:

1. Attempt all the questions and assume any suitable data, if necessary.
3. The Marks, CO (Course Outcome) and BL (Bloom's Level) related to questions are mentioned on the right-hand side margin.

1.	i. Differentiate between Reactive Agent model and Deliberative Agent model with neat diagram. ii. What is meant by Agent environment? Explain characteristics of agent environment.	5+5=10	CO1	Remember
2.	Explain AO* algorithm. What problems are associated with AO* algorithm? Solve the given problem using AO* algorithm where all numbers in brackets are the heuristic value i.e h(n). Each edge is considered to have a value of 1 by default. 	10	CO4	Analyze
3.	The tables below list the states in a search domain and heuristic values for each state, along with the transitions available from that state. For this question, assume that the start state is always **A** and the goal state is always **K**. A → B, C B → E C → D, G D → F E → I, J F → G → H, K H → L I → J → K K → L → A: 9 B: 4 C: 5 D: 7 E: 3 F: 10 G: 3 H: 5 I: 8 J: 2 K: 0 L: 7 Draw a diagram in the space that illustrates the search tree of explored states by indicating the calculated cost at each node in the tree, given an A* search. Calculate the minimum uniform path cost between states have to be for the heuristic to be admissible.	10	CO4, CO6	Analyze, Creative

4.	Suppose that we have data (with outcomes) S = {s1 = true, s2 = false, s3 = false, s4 = false} and one feature F that can have values {f1, f2, f3}. The feature value for s1 is f2, for s2 is f2, for s3 is f3 and for s4 is f1. Calculate information gain. F (feature) | S (outcome) S1->f2 | True S2->f2 | False S3->f3 | False S4->f1 | False	10	CO3	Apply
5.	(a) Translate the following sentences into First-Order Logic. i) Every student except George smiles. ii) Everyone walks or talks. iii) Every student who loves Mary is happy. iv) Every boy who loves Mary hates every other boy who Mary loves. v) Someone walks and someone talks. (b) Explain the Goal Stack Planning (GSP) method for the Block World problem with a suitable diagram and necessary functions. Also, explain partial order planning.	5+5=10	CO4	Analyze
6.	Consider the training dataset as shown in below table. Use this information to compute the conditional probabilities using Bayesian Classification for the Yes and No classes for the following new instance <sunny, cool, high, and true">.	10	CO4	Analyze

Outlook	Temp	Humidity	Windy	Play?
sunny	Hot	High	FALSE	No
sunny	Hot	High	TRUE	No
overcast	Hot	High	FALSE	Yes
rainy	mild	High	FALSE	Yes
rainy	cool	Normal	FALSE	Yes
rainy	cool	Normal	TRUE	No
overcast	cool	Normal	TRUE	Yes
sunny	mild	High	FALSE	No
sunny	cool	Normal	FALSE	Yes
rainy	mild	Normal	FALSE	Yes
sunny	mild	Normal	TRUE	Yes
overcast	mild	High	TRUE	Yes
overcast	Hot	Normal	FALSE	Yes
rainy	mild	High	TRUE	No

*****All the Best*****